

EXPLORATORY DATA ANALYSIS REPORT – NETFLIX DATASET

Introduction:

This report presents an Exploratory Data Analysis (EDA) of the Netflix Titles Dataset. The main objective of this analysis is to understand Netflix's content distribution, identify trends in movies and TV shows, clean the dataset, and generate meaningful business insights using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

Netflix is one of the world's leading streaming platforms, offering thousands of movies and TV shows across different genres, countries, and audience categories. By analyzing this dataset, we can understand Netflix's content strategy, audience targeting, and growth trends over time.

Objective of the Project:

The key objectives of this EDA project are:

- To understand the structure of the Netflix dataset
- To identify missing values and clean the data
- To analyze the distribution of Movies and TV Shows
- To identify top contributing countries
- To analyze trends in release years and content additions
- To study ratings and audience categories
- To explore movie duration and TV show seasons
- To identify the most popular genres on Netflix
- To generate insights using data visualizations

Dataset Overview:

Dataset Name

Netflix Titles Dataset 2021

Dataset Type

CSV File

Tools & Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Understanding

Initial exploration of the dataset was performed using the following functions:

```
# Display first rows
df1.head()

# Display last rows
df1.tail()

# Dataset information
df1.info()

# Statistical summary
df1.describe()
```

Observations

- Dataset contains both categorical and numerical features.
- Some columns contain missing values.
- Most missing values were found in:
 - director
 - cast
 - country
- The dataset includes both Movies and TV Shows.

Data Cleaning

Data cleaning is one of the most important steps in EDA. Missing values and incorrect data can affect analysis and visualization.

Missing Values Check

```
df1.isnull().sum()
```

Handling Missing Values

```
df1.dropna(subset=['date_added'], inplace=True)
```

Exploratory Data Analysis

Movies vs TV Shows Distribution

Objective:

To compare the number of Movies and TV Shows available on Netflix.

Visualization Used

Count Plot

Bar Chart

Insight

Netflix contains significantly more Movies than TV Shows.

Movies dominate the platform content library.

Sample Code

```
sns.countplot(x='type', data=df1)
plt.title('Movies vs TV Shows')
plt.show()
```

Country-wise Content

Analyzed which countries produce the most content

Insight

USA contributes the highest content Followed by countries like India, UK, etc.

Release Year Analysis

Studied trends in content release over time

Insight:

Significant growth after 2015

Peak content release around recent years

Ratings Analysis

Explored content ratings (e.g., TV-MA, PG, R)

Insight:

Most content is rated TV-MA (mature audience)

Duration Analysis

Movies: duration in minutes

TV Shows: number of seasons

Insight:

Most movies are around 90–120 minutes

Most TV shows have 1–2 seasons

Genre Analysis

Based on listed_in column

Insight:

Popular genres include:

Dramas

International Movies

Comedies

Visualization

Used:

Bar charts → Content distribution

Count plots → Ratings, countries

Histograms → Duration

Heatmaps → Correlation (if applied)

Key Insights

Netflix has a larger number of movies than TV shows

Content addition increased significantly after 2015

USA dominates content production

Majority content is targeted at mature audiences

Popular genres are Drama and International content

QUESTION

Q1. What is the objective of this EDA project?

Answer: The objective is to understand the Netflix dataset, clean missing values, and extract insights about movies and TV shows available on the platform.

Q2. What type of data does the dataset contain?

Answer: The dataset contains both –

- **Categorical data** (type, country, rating, genre)
- **Numerical data** (release_year , duration)

Q3. What are the key columns in the dataset?

Answer: Important columns include:

- Show_id
- type
- title
- director
- cast
- country
- date_added
- release_year
- rating
- duration
- listed_in (genre)

Q4. How did you handle missing values?

Answer:

- Checked using `isnull().sum()`
- Removed rows where `date_added` was missing using `dropna()`

Q5. Why was `date_added` important?

Answer:

It is important for **time-based analysis**, such as tracking content growth over years.

Q6. Which columns had the most missing values?

Answer:

- director
- cast
- country

7. Which type of content is more on Netflix?

Answer: Netflix has **more Movies than TV Shows**.

Q8. Which country produces the most Netflix content?

Answer: The **USA produces the highest amount of content**, followed by countries like India and the UK.

Q9. What trend was observed in release years?

Answer:

- Rapid growth after **2015**
- Peak content in recent years

Q10. What is the most common rating on Netflix?

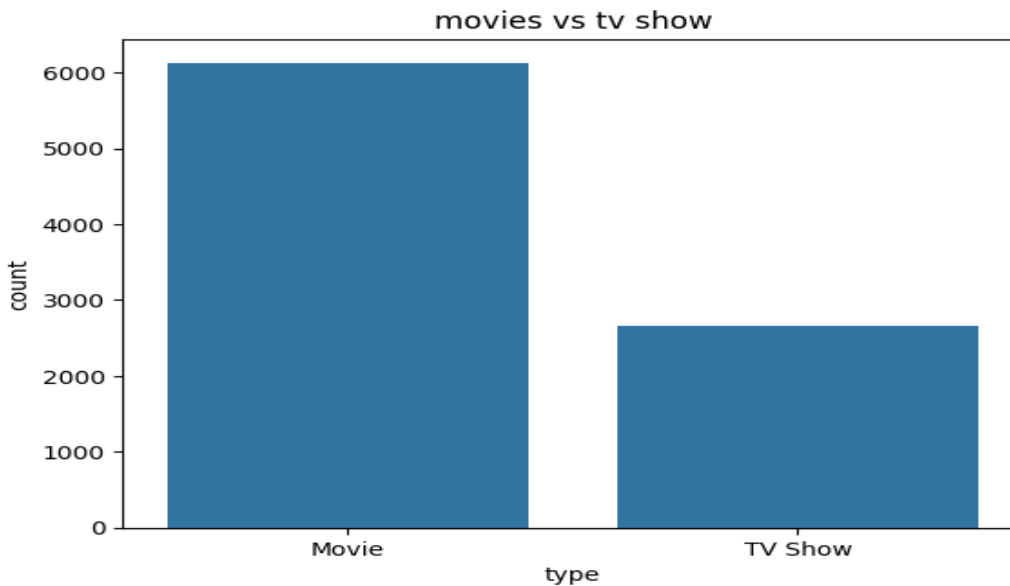
Answer: Most content is rated **TV-MA (Mature Audience)**.

Q11. What is the typical duration of movies?

Answer: Most movies are **90–120 minutes** long.

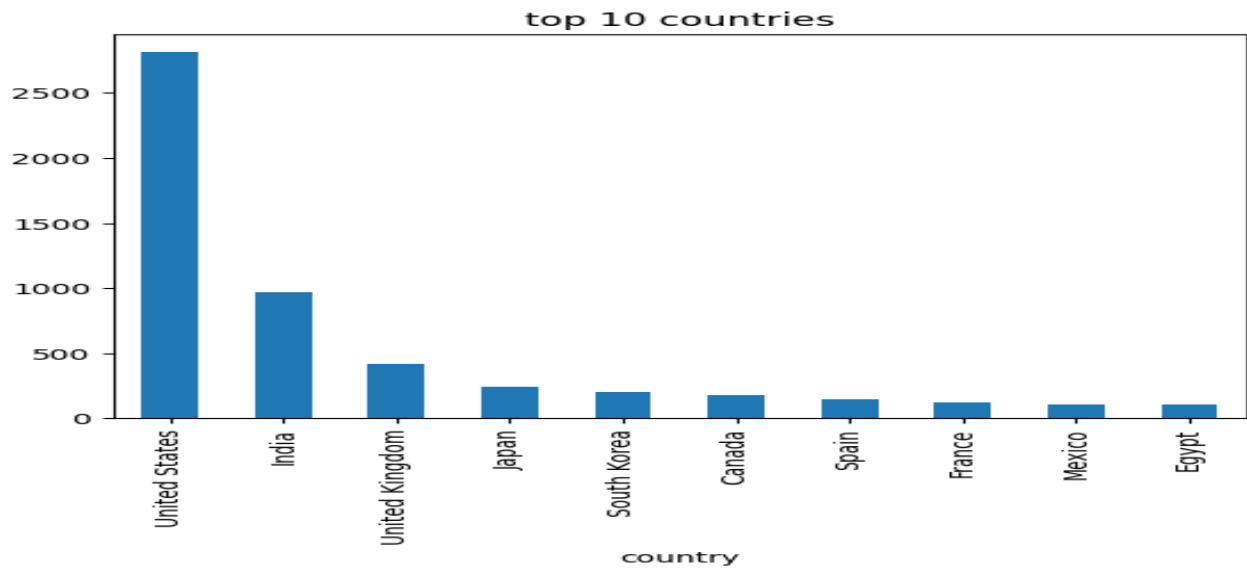
Q12. What does the Movies vs TV Shows graph show?

Answer: Movies dominate the platform compared to TV Shows.



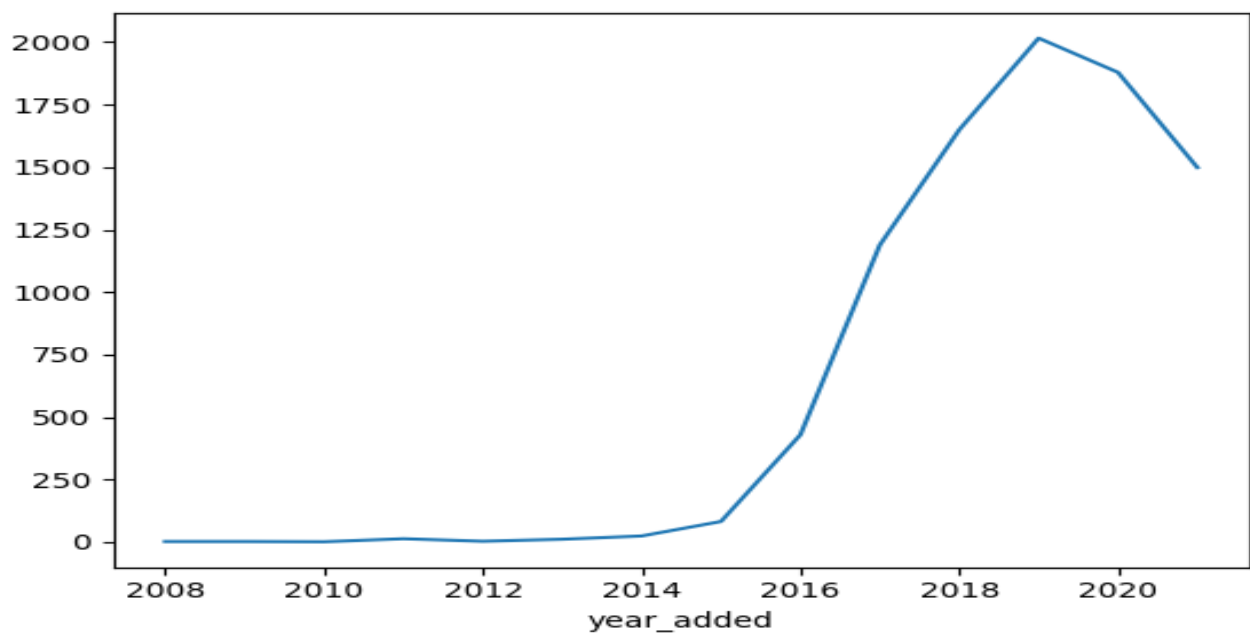
Q13. What insight do you get from the country graph?

Answer: USA produces the highest content, showing market dominance.



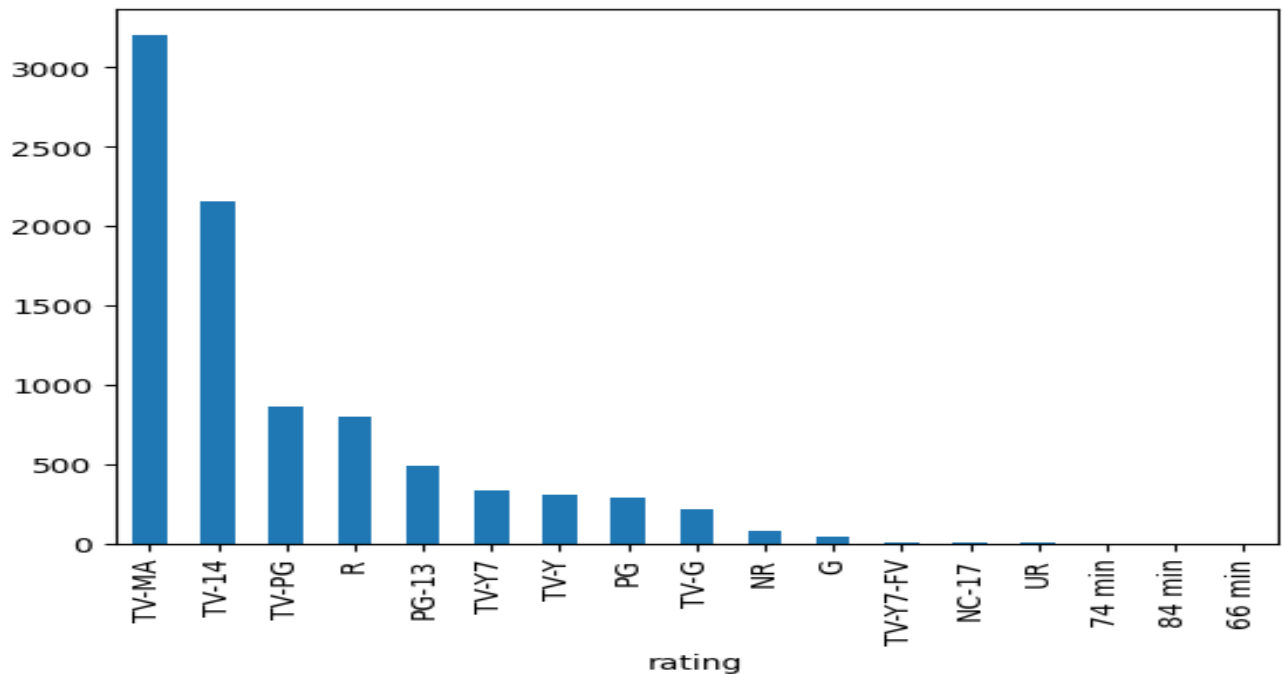
Q14. What trend is visible in the release year graph?

Answer: Content increased rapidly after 2015.



Q15. What does the rating graph indicate?

Answer: Most content is targeted at mature audiences (TV-MA).



Q16. What does the duration histogram show?

Answer: Most movies are between 90–120 minutes.

Q17. What are the key insights from the dataset?

Answer:

- Netflix has more movies than TV shows
- Content increased after 2015
- USA dominates content production
- Most content is for mature audiences
- Drama is the most popular genre

Business Insights

From the analysis, several business-level insights can be understood:

- Netflix invests heavily in movie-based content.
- International expansion is a major strategy.
- Mature audience content drives platform engagement.
- Genre diversity helps attract global subscribers.
- Rapid content growth after 2015 reflects increasing competition in streaming platforms.

Conclusion

- The Exploratory Data Analysis of the Netflix dataset provides meaningful insights into the platform's content strategy, growth, and audience targeting.
- The analysis reveals that Netflix primarily focuses on Movies rather than TV Shows. The dataset also shows rapid expansion in content after 2015, indicating strong growth in the streaming industry. The United States remains the largest contributor of Netflix content, followed by countries such as India and the United Kingdom.
- Most content falls under the TV-MA rating category, suggesting Netflix targets mature audiences more heavily. Popular genres such as Drama, International Movies, and Comedy dominate the platform, reflecting global viewer preferences.
- Despite missing values in certain columns, the dataset provides valuable insights after proper cleaning and preprocessing. This EDA can further support advanced analytics such as recommendation systems, audience behavior prediction, and content optimization.